Presenter	Title	Abstract
Yijing (Quinn) Lin	Priming Effects on Verbal Analogy Performance Across IQ and Working Memory in Humans and AI	Performance in verbal analogy tasks is well known to correlate with fluid intelligence, yet the underlying cognitive mechanisms remain unclear. We hypothesize that priming, prior exposure to word relationship, enhances analogy performance, and this enhancement is modulated by both IQ and age. Human participants completed a priming task followed by a set of verbal analogy problems, with intelligence independently assessed via Raven's Progressive Matrices. Our results show that priming significantly improves verbal analogy performance in individuals with lower IQ. However, among individuals with higher IQ, the priming effect is age-dependent: younger adults exhibit a reduced priming effect, whereas older adults show an enhanced effect. We propose that working memory capacity underlies these age-related differences. Specifically, with relatively higher working memory capacity, younger individuals are better able to integrate relational patterns from previous trials, thereby improving overall task performance while reducing susceptibility to priming. In contrast, older adults—with relatively lower working memory capacity—are more reliant on external cues provided by priming. We further simulated Large Language Models (LLMs) with configurations reflecting "high IQ" vs. "low IQ" and "large" vs. "small" working memory. The models exhibited similar patterns to those observed in human participants, supporting the proposed role of working memory capacity in modulating priming effects. These findings highlight the potential of LLMs as tools for investigating human cognitive performance in demanding tasks such as verbal analogy, including individual differences in fluid intelligence.
Zhino Ebrahimi	The Maze of Creative Thinking: Are We More or Less Creative in a Group?	Creativity is a fundamental aspect of human cognition, enabling adaptation, problem-solving, and innovation. While research predominantly focuses on individual creative performance, the influence of group dynamics on creativity remains underexplored. This study examines creativity in individual and team settings, evaluating whether collaborative efforts enhance or hinder creative outcomes. A total of 120 individuals participated in two sessions. The first session involved cognitive assessments, including the Advanced Progressive Matrices (APM), the Creative Reasoning Test (CRT), and the Test for Creative Thinking–Drawing Production (TCT-DP), as well as mood and personality questionnaires. We investigate the interplay between cognitive diversity, social influences, and theoretical models of team creativity, particularly the additive and disjunctive models. The additive model posits that group creativity emerges from the collective contributions of all members, whereas the disjunctive model suggests that team performance is driven by the most creative individual. Using an experimental design that systematically controls for cognitive diversity, we analyze creative outputs from both settings. Our findings reveal that while individuals often generate more original ideas, collaborative groups benefit from cognitive diversity, leading to enhanced idea refinement and synthesis. The results indicate that APM aligns with the disjunctive model, TCT-DP aligns with the additive model, while CRT does not fit either model. Keywords: Divergent Thinking, Convergent Thinking, Creative Reasoning Test (CRT), Creativity, Intelligence

Merima Halilbegović	Meaning vs. Form-Oriented Word Processing: Effects on Memory Traces	Previous research has explored the relationship between language meaning, form, and cognition, typically focusing on comprehension rather than memory. While meaning and form processing are often studied in more complex contexts like text comprehension, single-word processing has mainly been examined through priming effects. However, few studies have directly investigated how isolated word processing influences memory retention. This study explores how emphasis on meaning versus form in word processing affects memory retention. We hypothesized that semantic processing would lead to stronger memory traces than form-based processing. To test this, 96 participants completed an online questionnaire consisting of 10 multiple-choice questions, each with four alternative noun answers. Half of the questions directed participants to focus on form, while the other half emphasized meaning. The questions were counterbalanced to control for order effects. After completing the questionnaire, participants were asked to recall as many of the 40 nouns presented as alternatives across all questions, without prior knowledge of the recall task. Results showed that participants recalled significantly more words from the meaning-focused questions, suggesting that semantic processing enhances memory retention more effectively than form-based processing. Several limitations should be noted, including the small sample size, reliance on a single task format, and lack of control over noun type. Future research could expand on these findings by exploring a wider range of word types, task designs, and participant demographics to better understand the role of meaning-based processing in memory across various contexts. This study provides more insight into the impact of meaning-based versus form-based processing on memory retention, with potential applications in educational, marketing, and other cognitive fields.
Hening Wang	Interpreting Plurals in Context: Effects of Visual Priming and NP Form on Cover Selection	Plural predication in natural language is often ambiguous, allowing for multiple interpretations depending on context. A sentence such as “The Xs are tall” can refer to each member of the group individually (distributive reading) or to the group as a whole (collective reading). Moreover, plural statements may support intermediate interpretations, in which subsets within a group contribute differently to the property in question. This project investigates how visual context and nominal phrase (NP) structure interact to influence the interpretation of plural sentences. Specifically, we test whether visual priming can increase the salience of certain “cover” contexts—ways of grouping individuals—to bias interpretation, and whether linguistic structures (e.g., Xs vs. As and Bs) influences this effect. By systematically manipulating visual displays and NP structure, we found that while conjunctions are primarily shaped by the corresponding visual context, definite plurals support a broader range of cover selections.
Sia Ivova Hranova	Structure learning in a non-parametric Bayesian Contextual Control Model	Contextual inference plays a substantial role in the creation, expression, and updating of memories in animals and humans, with experimental evidence coming from fields ranging from motor learning (Heald, Lengyel & Wolpert., 2021; Cuevas Rivera & Kiebel, 2023) to value-based decision making (Collins & Koechlin, 2012). To unify the resulting disparate definitions of context, contextual processing can be expressed as probabilistic inference, where memory management is contingent on continually inferring the current context, and context is defined as a discrete top-level variable that “indexes” memories (Heald, Lengyel & Wolpert, 2023). The recently proposed Active Inference Bayesian Contextual Control (BCC) model (Schwöbel et al., 2021) employs this framework to learn context-dependent environmental contingencies and behavioral heuristics via free energy minimization. Here, our goal is to generalize the BCC model to account for context-dependent memory creation in addition to memory update and expression. This will enable the model to perform structure learning by adapting contextual representation complexity to the current task. To achieve this, we introduced a non-parametric Dirichlet process (DP) prior (Teh, 2010) over contexts. We derived and compared various DP-based extensions of the BCC model and characterized their ability to perform stable online inference in a multiarmed bandit task. Simulations highlight the importance of prior beliefs over context contingencies for performance and raise key research questions such as how newly opened contexts are initialized. The resulting model enables investigating how humans adaptively learn contextual representations in an uncertain and changing environment in a principled manner. Affiliation: Chair of Cognitive Computational Neuroscience, Faculty of Psychology, Dresden University of Technology, Dresden, Germany Authors: Sia Ivova Hranova, Stefan Kiebel, Sarah Schwöbel

Ekaterina Varkentin	Visual Narrative Comprehension in the Context of Aging and Stress	Visual narrative comprehension is crucial in today's visually driven world, yet research on this ability in the context of aging remains inconclusive. Further, the interplay between aging and stress in narrative comprehension remains largely unexplored, even with stress becoming an ever-present factor in modern life. This online study examined the effects of acute stress on visual narrative comprehension in younger (N = 203, 18-57 years) and older adults (N = 212, 60-85 years). Participants were assessed under both acute stress and neutral conditions. An online stress induction tool combined mathematical and logical tasks under time pressure with elements designed to induce social stress. Participants were presented with pictorial stories comprising three panels, where the second panel was intentionally left blank. On the next page, participants judged whether the given inference for the blank panel was correct. Results showed that acute stress impaired visual comprehension and confidence in their answer in younger adults, while older adults remained unaffected. These findings suggest that with age and experience, individuals develop more differentiated event schemas, enhancing resilience to stress. The new insights provide a deeper understanding of visual narrative comprehension in the context of aging and acute stress.
Pritha Sen	Identifying Early Cognitive and Behavioral Alterations in Psychosis Risk Using Computational Modeling	Aberrant reward processing and cognitive control deficits predict psychosis risk, emerging as early indicators of vulnerability. This study examined whether computationally derived markers differ in children with high psychotic-like experiences (PLEs) from controls. From the ABCD data, we identified a high PLE group (N=131) based on Prodromal Psychosis Syndrome (PPS) score>3 and distress score>6, and matched to controls at baseline and 2nd-year follow-up (N=131). We analysed response vigor in the monetary incentive delay (MID) task using generative modeling, and estimating intercept, cue salience, novelty, post-error adjustment, and linear trend. N-back and stop-signal tasks (SST) were analyzed using drift diffusion modeling to estimate drift rate, boundary separation, and non-decision time. SST performance was also evaluated using stop-signal reaction times (SSRTs). Additionally, we conducted correlations at both time-points in the full sample (N=6457) between model parameters, PPS, and distress scores. MID: At follow-up, the high PLE group showed lower cue salience, suggesting altered reward sensitivity. Correlations at baseline showed that higher PPS and distress were linked to increased intercept and linear trend, reflecting heightened response tendencies. n-back: the high PLE group had lower drift rates and increased boundary separation at both time points, indicating inefficiencies in evidence accumulation and cautious responses. Lower drift rates at baseline correlated with higher PPS and distress. SST: the high PLE group had longer SSRTs at baseline, reflecting impaired inhibitory control. At follow-up, slower drift rates and longer non-decision time indicated inefficient inhibitory control and processing delays. These preliminary results suggest that reward processing deficits might emerge later, while cognitive control impairments persist, highlighting the potential of computational modeling to identify neurocognitive vulnerabilities in psychosis risk.

Mayte H. Laureano	Computational Study of Dream Interpretations: Psychoanalytic Human vs Artificial Analyses	We compare how human analysts and ChatGPT-4 interpret dreams, using lexical and semantic categories defined by the LIWC tool. Our dataset includes interpretations from 38 analysts for 20 patients, allowing us to identify differences between human and AI-generated analyses. ChatGPT-4 focuses more on meaning (semantic categories) rather than grammar, especially favoring words related to vision and health. We use Naïve Bayes classification to distinguish between human and AI interpretations, achieving high accuracy. Our findings show that ChatGPT-4 does not consistently align with the central pattern of human analysis but often differs in its narrative style. This study contributes to the discussion on AI's role in dream interpretation, suggesting that while ChatGPT-4 creates clear and structured narratives, it analyzes dreams in a way that differs from humans. More research is needed to explore how AI can be integrated into psychological practice.
rabia Dilawar	Fast and Focused: Exploring If-Then Plans in Visual Identification and Cognitive Control of Food Stimuli	Cognitive control plays a crucial role in regulating eating behaviors, particularly in an environment abundant with food cues .Implementation intentions (If-Then Plans) have been shown to facilitate self-regulation through cue-response association by creating an effective pathway toward goal-directed .This study investigates the differential effect of implementation intentions (If-Then Plans) compared with standard goal intentions on cognitive control and decision-making when people are exposed to food-related stimuli. A total of 79 participants completed Classification tasks across the three conditions: baseline, cued (If-Then Plan), and self-initiated (Goal Intention). Participants were randomly assigned into two groups according to different statement conditions; each group went through all three counterbalanced conditions. The primary outcome measures included reaction times, error rates, and Dutch Eating Behavior Questionnaire scores (administered before the experiment). A linear mixed-effects model showed that the If-Then Plan and Goal Intention conditions' reaction times were significantly faster than the baseline. In Group 1, If-Then Plans indeed produced a slightly greater reduction compared with goal intentions, $M = 713.32$ and 714.24 ms (βs $= -47.43$ and -46.51, ps $<.001$), respectively, from baseline, $M = 760.75$ ms. Goal Intentions ($M = 716.08$ ms, $\beta = -38.79$, $p <.001$) had a slightly larger effect size than If-Then Plans did ($M = 719.81$ ms, $\beta = -35.05$, $p <.001$) relative to the baseline ($M = 754.86$ ms) in Group 2. Random effects analysis revealed large individual differences: $\sigma^2\_ID = 1642–1430$, $\sigma^2\_residual = 6508–6819$. The findings indicate that if-then planning strategies enhance cognitive control in food-related decision-making. Future studies should further investigate their longitudinal effects on eating behavior and moderation of their impact by individual-level factors, like habit strength and motivation. Keywords: If-then plan, Goal intention, eating disorder, self-regulation, Food cues.
Annika Oldach	The 'I' in Narrative: How Reader Characteristics Influence the Effect of Narrative Texts on Text Comprehension	In this thesis, we investigate the impact of text genre (expository vs. informative narrative) on comprehension and examine indirect and interaction effects of text-related beliefs and certain reader characteristics, respectively. The primary research question addresses whether need for cognition (NFC) and reading skill moderate the effects of text genre on text-related beliefs and comprehension. The results indicate that NFC does not significantly moderate the relationship between text genre and factual comprehension, but shows a trend towards influencing inferential comprehension, where individuals with a high NFC show worse comprehension for informative narratives. Reading skill was found to affect comprehension in such a way that informative narratives were more detrimental to inferential comprehension for less proficient readers. In contrast, well-developed reading skills seem to protect against the negative effect of informative narratives, at least to some degree. The study highlights the need to tailor educational texts to individual reader characteristics and suggests that while expository texts generally support better comprehension, narratives may complicate understanding, particularly for less skilled readers. Limitations include a small sample size and the exclusion of prior knowledge as a variable. Future research should explore additional reader characteristics, such as interest, motivation, or working memory capacity, and include prior knowledge to further explain these findings.

Juan Peschken	Context is Learned, not Given	The saying "context is everything" underscores the importance of interpreting things, be they quotes, events, actions, or stimuli, not in isolation but in the light of a bigger picture - their context. This is evident even in fundamental forms of learning such as extinction learning where, in contextual renewal, an extinguished response reoccurs if the context is changed. But what exactly is context? Is context given by stimuli with inherent properties making them context or, what are the circumstances that allow a stimulus to become "contextual"? Even though the answer may seem intuitively trivial, the literature only provides competing and vague definitions. Using a novel paradigm in pigeons, we directly assessed competing stimuli in their ability to induce contextual renewal during extinction learning. Furthermore, we controlled the timing of these stimuli and found it to be crucial; with the right contiguity, even small local stimuli resulted in the strongest contextual renewal. This result challenges definitions of context as 'a backdrop where learning occurs'. Instead, we propose that context can be understood mechanistically as a learned stimulus property. Therefore, context truly is everything and anything.
Onur Icin	A Computational Framework for Multimodal Biomarkers in Early Psychosis Risk	Identification of psychosis risk at its earliest stages, particularly in adolescents experiencing psychotic-like experiences (PLEs), remains a critical challenge in clinical psychiatry due to the dynamic nature of neurodevelopment during this period, which can make early risk difficult to define. In our current work, we have implemented a computational framework that integrates structural imaging data from voxel-based morphometry (VBM) and diffusion-weighted imaging (DWI) with key cognitive test scores and clinical assessments based on psychosis risk measures, including subclinical scores of PLEs. This multimodal approach enables us to investigate the individual and combined contributions of neuroanatomical structure, white matter connectivity, and cognitive performance in distinguishing adolescents at increased risk for psychosis. Using the ABCD data, we employ two time-points to assess the longitudinal prediction of psychosis risk. Using binary classification, our preliminary analyses reveal that both imaging and cognitive measures contribute unique predictive information. Preliminary results indicate that imaging markers when combined with cognitive performance in a multimodal approach, enhance risk stratification compared to unimodal approaches. Notably, the combined model outperformed unimodal models by approximately 10–12%, suggesting that cognitive performance adds substantial predictive value beyond neuroimaging alone. The insights support the potential of this computational framework to advance early psychosis prediction. Our initial results show that integrating structural imaging markers and cognitive information provides a more reliable picture for psychosis risk. This combined approach could help lay the groundwork for more targeted and effective early intervention strategies in clinical psychiatry.
Jens Kreber	Evaluating the Effect of Noise on a Probabilistic Robot- and Terrain-Aware Dynamics Model	Mobile robots should be capable of planning cost-efficient paths, even if terrain properties, such as friction, vary with location. TRADYN is a terrain- and robot-aware forward dynamics model for a simulated 2D unicycle robot, which is conditioned on a context variable and terrain feature lookups, and can therefore adapt to various environment properties. We evaluate the prediction and planning performance of TRADYN under different sources of noise. We find that terrain lookup and especially calibration have a positive effect on prediction performance even under noisy conditions. A preprint including our analysis can be found at https://arxiv.org/abs/2409.11452

Yuxin Lu	The realization of tones in spontaneous spoken Taiwan Mandarin: a corpus-based survey and theory-driven computational modeling	A growing body of literature has demonstrated that semantics can co-determine fine phonetic details. For instance, Chuang et al. (2024) found that the pitch contours of Mandarin disyllabic words with a T2-T4 tone pattern can be predicted from their semantics with above-chance accuracy. The current study investigates the pitch realization of disyllabic words with 20 bi-tonal combinations using a corpus of spoken Taiwan Mandarin (Fon, 2004). We first used Generalized Additive Mixed Models (GAMs; Wood, 2017) to model f0 contours as a function of predictors, including gender, neighboring tones, tone pattern, speech rate, word position, bigram probability, speaker, word, and words' meaning. In the GAM analysis, word and words' meaning emerged as crucial predictors of f0 contours, with effect sizes exceeding those of tone pattern. This findings replicated the word-specific tonal realization previously reported by Chuang et al. (2024) and Lu et al. (2024). Furthermore, we mapped the semantic vectors, represented by 768-dimensional Contextualized Embeddings, onto these pitch contours using Linear Discriminative Learning (Baayen et al., 2019). We implemented two methods for calculating the pitch vector. The mean prediction accuracy was 2.8% for the training dataset and 1.4% for the testing dataset. Both accuracies were above the permutation baseline of 0.4%. Moreover, we predicted the pitch realization of 20 tone patterns from the centroid of Contextualized Embeddings for the tone patterns. The LDL-predicted pitch realization of tone patterns showed considerable similarity with the GAM-derived contours (see Figure 1). Our results support two conclusions. First, words and their meanings co-determine the realization of the f0 contours in disyllabic words, with a greater effect than tone pattern. Second, words' pitch contours can be predicted to a considerable extent from their meanings in context. This finding suggests that the mapping from context-sensitive meaning to pitch contours is machine-learnable, providing strong support for the possibility that human learners can also achieve this, given the dynamic and flexible nature of the mental lexicon (Libben, 2022). References Baayen, R. H., Chuang, Y.-Y., Shafaei-Bajestan, E., & Blevins, J. P. (2019). The discriminative lexicon: A unified computational model for the lexicon and lexical processing in comprehension and production grounded not in (de) composition but in linear discriminative learning. Complexity, 2019(1), 4895891. https://doi.org/10.1155/2019/4895891 Chuang, Y.-Y., Bell, M. J., Tseng, Y.-H., & Baayen, R. H. (2024). Word-specific tonal realizations in Mandarin. arXiv preprint arXiv:2405.07006. https://doi.org/10.48550/arXiv.2405.07006 Fon, J. (2004). A preliminary construction of Taiwan Southern Min spontaneous speech corpus. Technical report, Tech. Rep. NSC-92-2411-H-003-050, National Science Council, Taiwan. Libben, G. (2022). From lexicon to flexicon: The principles of morphological transcendence and lexical superstates in the characterization of words in the mind. Frontiers in Artificial Intelligence, 4, 788430. https://doi.org/10.3389/frai.2021.788430
Ruiqi He	Experience-driven discovery of planning strategies	One explanation for how people can plan efficiently despite limited cognitive resources is that we possess a set of adaptive planning strategies and know when and how to use them. But how are these strategies acquired? While previous research has studied how individuals learn to choose among existing strategies, little is known about the process of forming new planning strategies. In this work, we propose that new planning strategies are discovered through metacognitive reinforcement learning. To test this, we designed a novel experiment to investigate the discovery of new planning strategies. We then presented metacognitive reinforcement learning models and demonstrated their capability for strategy discovery as well as show that they provided a better explanation of human strategy discovery than alternative learning mechanisms. However, when fitted to human data, these models exhibited a slower discovery rate than humans, leaving room for improvement.
Paul Hege	Probing neural representations of semantic relations with an analogy task	TBD.

Hassane Kissane	Neural and computational Modeling of Verb-Particle combinations	TBD.
Joseph Petrie	Weak assertion: a decision-theoretic rational speech acts model	TBD.